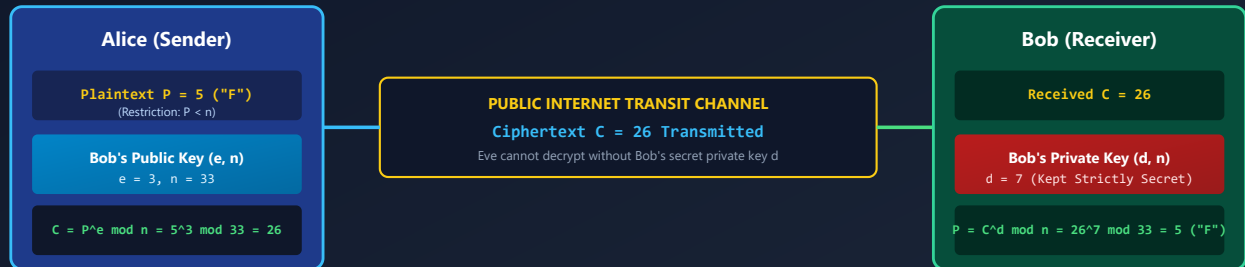


Module 11: Asymmetric Cryptography and RSA Algorithm

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ASYMMETRIC CRYPTOGRAPHY & RSA ALGORITHM

Public-Key Mechanics, Prime Modulus Math, Key Generation Steps & Solved AKTU Example



RSA ALGORITHM STEP-BY-STEP MATHEMATICAL FOUNDATIONS (SLIDES 78 - 82)

1. Key Generation Steps

- 1. Choose 2 large primes:

p and **q**. • 2. Compute modulus: $n = p \times q$. • 3. Compute Euler phi: $\phi(n) = (p-1)(q-1)$. • 4. Choose **e**: $1 < e < \phi(n)$ & $\gcd(e, \phi(n)) = 1$. • 5. Compute **d**: $d \times e \equiv 1 \pmod{\phi(n)}$. 2. Encryption & Decryption • **Public Key**: (e, n) | **Private Key**: (d, n) • **Encryption**: $C = P^e \bmod n$ • **Decryption**: $P = C^d \bmod n$ • **Security Foundation**: Integer Factorization Factoring large n (~2048 bits) is intractable. 3. Solved Numerical Example • $p = 3, q = 11 \rightarrow n = 33, \phi(n) = 20$ • Choose $e = 3$ ($\gcd(3, 20) = 1$) • $d: 3d \equiv 1 \pmod{20} \rightarrow 3 \times 7 = 21 \equiv 1 \rightarrow d = 7$ • Encrypt $P=5$: $C = 5^3 \bmod 33 = 125 \bmod 33 = 26$ • Decrypt $C=26$: $P = 26^7 \bmod 33 = 5$ ("F")

1. Asymmetric Public-Key Cryptography

Asymmetric cryptography eliminates key exchange vulnerability by deploying dual mathematically coupled keys: a universally published Public Key and a confidential Private Key.

The Fundamental RSA Mathematical Engine

Security rests upon the computational difficulty of factoring the product $n = p \times q$ of two massive prime integers.

2. RSA Key Selection Sequence

1. Select distinct large primes p and q .
2. Calculate modulus: $n = p \times q$.
3. Compute totient: $\phi(n) = (p - 1)(q - 1)$.
4. Choose public exponent e such that $\gcd(e, \phi(n)) = 1$.
5. Compute private exponent $d \equiv e^{-1} \pmod{\phi(n)}$.

Encryption: $C = P^e \bmod n$ | Decryption: $P = C^d \bmod n$ ($P < n$)

3. AKTU Solved Numerical: Transmission of "F"

Parameter	Value	Calculation Rule
Prime Inputs	$p = 3, q = 11$	Given prime parameters
Modulus & Totient	$n = 33, \phi(n) = 20$	$n = 3 \times 11, \phi(n) = 2 \times 10 = 20$
Public & Private Exponents	$e = 3, d = 7$	$\gcd(3, 20) = 1, 3 \times 7 = 21 \equiv 1 \pmod{20}$

Plaintext "F"	$P = 5$	0-indexed alphabet code (A=0, ..., F=5)
Ciphertext Result	$C = 26$	$5^3 \bmod 33 = 125 \bmod 33 = 26$
Decrypted Plaintext	$P = 5$ ("F")	$26^7 \bmod 33 = 5$